Project: Worked examples
Title: Tolerance stack-up (1D)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

A linear tolerance loop: worst-case limits, root-sum-square and modified RSS limits, predicted spread and capability against a stated requirement, contribution Pareto and a Monte Carlo prediction.

WHAT IT CALCULATES	RELATION
Worst case, every contributor at its limit	$T_{wc} = \Sigma\lvert\text{half-bands}\rvert$
Root sum of squares, the statistical stack	$T_{rss} = \sqrt{\Sigma}\ \text{half}^2$
Modified RSS, a factor on the RSS (Bender-style)	$T_{mrss} = \text{factor} \times \text{RSS}$
Predicted standard deviation of the stack, from each contributor's distribution	$\sigma = \sqrt{\Sigma}\ \sigma_i^2$
Process capability, taken from the nearer limit	$C_{pk} = \min(\text{USL} - \mu, \mu - \text{LSL})/(3\sigma)$
Monte Carlo check, every contributor sampled from a fixed seed and summed with its sign	$\text{stack} = \Sigma\ s_i\ x_i$

Basis: Technical product documentation and specification[1]; Dimensioning and tolerancing[2]; General tolerances (where a contributor's band is a general tolerance)[3].

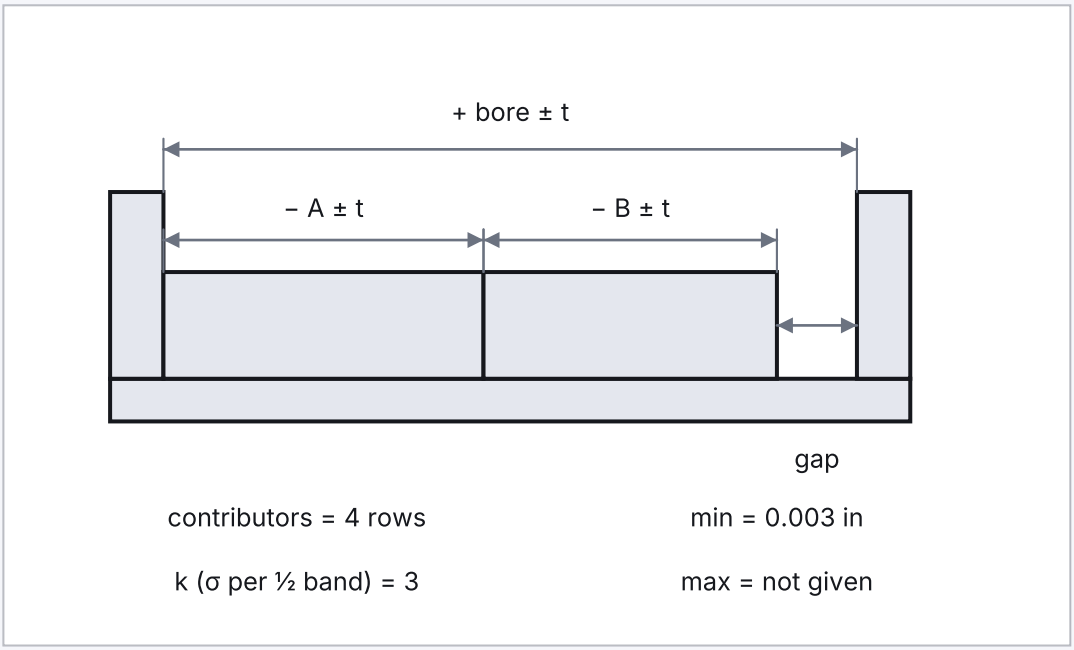


Figure 1 — A linear stack (representative; the contributors as listed): the bore adds, the parts subtract, the gap closes the loop. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
contributors	Contributors in the loop	4 row(s)		spec: Shigley 9e Ch. 1 example, part dimensions	any
minimum	Minimum acceptable result	0.003	in	spec: Shigley 9e Ch. 1 example, required gap	-39370 … 39370 in
k	Standard deviations per half-band (normal contributors)	3		assumption: Each part at ±3σ within its band.	1 … 6
f_{rss}	Modified-RSS factor (Bender-style)	1.5		assumption: Bender factor 1.5.	1 … 2
cpk_{target}	Required C_{pk}	1.33		assumption: Usual capability target.	1 … 2
n_{mc}	Monte Carlo samples	10000		assumption: Default sample count.	1000 … 100000
seed	Monte Carlo seed	1		assumption: Fixed seed for reproducibility.	0 … 4295000000
notation	Drawing notation	ASME Y14.5		assumption: Inch drawing.	any

TABLE ROW	VALUES
contributors[1]	name = a shoulder length; direction = adds; nominal = 1.75 in; upper = 0.003 in; lower = -0.003 in
contributors[2]	name = b sleeve; direction = subtracts; nominal = 0.75 in; upper = 0.001 in; lower = -0.001 in
contributors[3]	name = c washer; direction = subtracts; nominal = 0.12 in; upper = 0.005 in; lower = -0.005 in
contributors[4]	name = d sleeve; direction = subtracts; nominal = 0.875 in; upper = 0.001 in; lower = -0.001 in

3. Assumptions

A1. Every contributor acts along the one axis of the stack. (A linear stack: no angular or datum-shift effects beyond what the user enters as a contributor.)
A2. Statistical figures assume contributors vary independently about their mid-limit (plus any stated mean shift), each with the distribution stated for it. (The worst case is a guarantee; every statistical figure is a prediction about a population.)
A3. A position or profile contributor contributes ± half its tolerance zone, plus half the stated MMC bonus. (Diametral zones are bilateral about the true position.)
A4. k (Standard deviations per half-band (normal contributors)) is taken as 3. (Each part at ±3σ within its band.)
A5. f_{rss} (Modified-RSS factor (Bender-style)) is taken as 1.5. (Bender factor 1.5.)
A6. cpk_{target} (Required C_{pk}) is taken as 1.33. (Usual capability target.)
A7. n_{mc} (Monte Carlo samples) is taken as 10000. (Default sample count.)
A8. seed (Monte Carlo seed) is taken as 1. (Fixed seed for reproducibility.)

A9. notation (Drawing notation) is taken as ASME Y14.5. (Inch drawing.)

4. Working

[6] c1 = + 1.75 +0.003/-0.003
= mid 1.75 in, ± 0.003 in, σ 0.001 in (normal)
= 1.75 in
a shoulder length

[7] c2 = − 0.75 +0.001/-0.001
= mid 0.75 in, ± 0.001 in, σ 0.000333333 in (normal)
= -0.75 in
b sleeve

[8] c3 = − 0.12 +0.005/-0.005
= mid 0.12 in, ± 0.005 in, σ 0.00166667 in (normal)
= -0.12 in
c washer

[9] c4 = − 0.875 +0.001/-0.001
= mid 0.875 in, ± 0.001 in, σ 0.000333333 in (normal)
= -0.875 in
d sleeve

[10] x_{nom} = Σ (sign × nominal)
= 1.75 − 0.75 − 0.12 − 0.875
= 0.005 in
Nominal result

[11] x_{mean} = Σ (sign × mid-limit)
= 1.75 − 0.75 − 0.12 − 0.875
= 0.005 in
Mean result

[12] t_{wc} = Σ half-bands
= 0.003 + 0.001 + 0.005 + 0.001
= 0.01 in
Worst-case tolerance (±)

[13] $x_{wc,min}$ = x_{mean} − t_{wc}
= 0.005 in − 0.01 in
= -0.005 in
Worst-case minimum

[14] $x_{wc,max}$ = x_{mean} + t_{wc}
= 0.005 in + 0.01 in
= 0.015 in
Worst-case maximum

[15] t_{rss} = √Σ half²
= √(0.003² + 0.001² + 0.005² + 0.001²)
= 0.006 in
Root-sum-square tolerance (±)

[16] $x_{rss,min}$ = x_{mean} − t_{rss}
= 0.005 in − 0.006 in
= -0.001 in
RSS minimum

[17] $x_{rss,max}$ = x_{mean} + t_{rss}
= 0.005 in + 0.006 in
= 0.011 in
RSS maximum

[18] t_{mrss} = f_{rss} * t_{rss}
= 1.5 × 0.006 in
= 0.009 in
Modified RSS tolerance (±)

[19] $\sigma = \sqrt{\Sigma\ \sigma_i^2}$
$= \sqrt{(0.001^2 + 0.000333333^2 + 0.00166667^2 + 0.000333333^2)}$
$= 0.002$ in
Predicted standard deviation of the result
[20] $m_{min} = x_{wc,min}$ - minimum
$= -0.005 - 0.003$
$= -0.008$ in
Worst-case margin to the minimum
[21] $cpk = \min(x_{mean} - LSL, USL - x_{mean}) / (3\ \sigma)$
$= \min(0.002, \infty) / (3 \times 0.002)$
$= 0.33333$
Process capability index C_{pk}
[22] $dpmo = 10^6 \times P(outside)$
$= 10^6 \times \Phi\text{-tails at } \sigma = 0.002$
$= 158660$
Predicted defects per million
[23] $yield = 1 - dpmo / 10^6$
$= 1 - 158660 / 10^6$
$= 84.134$ %
Predicted yield
[24] $u_{cpk} = cpk_{target} / cpk$
$= 1.33 / 0.33333$
$= 3.99$
Capability utilisation
[25] pareto = largest contributor (statistical)
= a shoulder length 25 %; b sleeve 2.78 %; c washer 69.4 %; d sleeve 2.78 %
= "c washer"
Contribution Pareto
[28] mc_{mean} = mean of 10000 samples (seed 1)
= Monte Carlo
= 0.0050019 in
Monte Carlo mean
[29] mc_{sigma} = sample standard deviation
= Monte Carlo
= 0.0019919 in
Monte Carlo σ
[30] mc_{p05} = 5th percentile
= Monte Carlo
= 0.0017324 in
Monte Carlo 5 %
[31] mc_{p95} = 95th percentile
= Monte Carlo
= 0.0082769 in
Monte Carlo 95 %
[32] mc_{yield} = samples inside the requirement
= 8461 of 10000
= 84.61 %
Monte Carlo yield

5. Results

SYMBOL	VALUE	DESCRIPTION
x_{nom}	0.0050 in	Nominal result
x_{mean}	0.0050 in	Mean result

SYMBOL	VALUE	DESCRIPTION
t_{wc}	0.010 in	Worst-case tolerance (±)
$x_{wc,min}$	-0.0050 in	Worst-case minimum
$x_{wc,max}$	0.015 in	Worst-case maximum
t_{rss}	0.0060 in	RSS tolerance (±)
t_{mrss}	0.0090 in	Modified RSS tolerance (±)
σ	0.0020 in	Predicted σ
m_{min}	-0.0080 in	Worst-case margin to the minimum
cpk	0.33333	C_{pk}
dpmo	158660	DPMO (predicted)
yield	84.1 %	Yield (predicted)
mc_{mean}	0.0050 in	Monte Carlo mean
mc_{sigma}	0.0020 in	Monte Carlo σ
mc_{p05}	0.0017 in	Monte Carlo 5th percentile
mc_{p95}	0.0083 in	Monte Carlo 95th percentile
mc_{yield}	84.6 %	Monte Carlo yield

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Worst case within requirement	-0.005 … 0.015 in	0.003 … open in	—	
FAIL	C_{pk} against target	u_{cpk} = 3.99	u_{cpk} ≤ 1	3.99	

7. Applicability

None.

8. Verification

Template mech.tolerance-stack@2.2.1 Content hash 6ec171c7c89dab1e Engine calc-engine/1.0.0
Known-answer cases passed 7/7 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution. Technical Product Documentation and Specification. BS 8888. BSI.
2. American Society of Mechanical Engineers. Dimensioning and Tolerancing. ASME Y14.5. ASME.
3. International Organization for Standardization. General Tolerances—Part 1: Tolerances for Linear and Angular Dimensions Without Individual Tolerance Indications. ISO 2768-1. ISO.